PHY202 Practice Exam on chapter(s): 22

Name:____

These questions assess your understanding of the material from Chapter(s) 22 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. A Hall-effect flow probe is placed on an artery, applying a $0.36\,\mathrm{T}$ magnetic field across it. What is the Hall emf, given the vessel's inside diameter is $2.6\,\mathrm{mm}$ and the average velocity of the blood is $86.\,\mathrm{cm/s}$?
(Chapter 22, Section (6) The Hall Effect - cp1e_ex22_3)

ANSWER: ____________________________

2. A wire carries a 170 A current toward the south. If a large magnet were placed directly to the west of the wire and emits a $2.8\,\mathrm{T}$ magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.
(Chapter 22, Section (7) Magnetic Force on a Current-Carrying Conductor - cp1e_ex22_4)

ANSWER: ____________________________

3. An electron travels with speed $2.9 \times 10^6\,\mathrm{m/s}$ at an angle of 17.□° to a uniform magnetic field. If its path has a circular component of radius $0.061\,\mathrm{m}$, calculate the magnitude of the magnetic field.
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - tkd1)

ANSWER: ____________________________

4. The hot and neutral wires supplying DC power to a light-rail commuter train carry $970\,\mathrm{A}$ and are separated by $41.\,\mathrm{cm}$. What is the magnitude of the force between $70.\,\mathrm{m}$ of these wires?
(Chapter 22, Section (10) Magnetic Force betewen Two Parallel Conductors - cp1e_22_#50)

ANSWER: ____________________________

5. Find the maximum torque on a $22.$-turn square loop of wire that is $11.\,\mathrm{cm}$ on each side that carries $8.0\,\mathrm{A}$ of current in a 0.52-T field.
(Chapter 22, Section (8) Torque on a Current Loop: Motors and Meters - cp1e_ex22_5)

ANSWER: ____________________________

6. An electron moves at $3.3 \times 10^6 \frac{\mathrm{m}}{\mathrm{s}}$ perpendicular to a magnetic field. The field causes the proton to travel in a circular path of radius $44.\,\mathrm{cm}$. What is the strength of the magnetic field?
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_22_#13)

ANSWER: ____________________________

7. Suppose that you rub a glass rod with silk, placing a $69.$-nC positive charge on it. Calculate the initial force on the rod due to the Earth's magnetic field, if you throw the rod with a horizontal velocity of $2.4\,\mathrm{m/s}$ due west in a place where Earth's field is due north. Recall that the magnitude of Earth's magnetic field is $46.\,\mu\mathrm{T}$.
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ex22_1)

ANSWER: ____________________________

8. A large water main is $1.1\,\mathrm{m}$ in diameter and the average water velocity is $0.32\,\frac{\mathrm{m}}{\mathrm{s}}$. Find the Hall voltage produced if the pipe runs perpendicular to Earth's 4.9×10^{-5}-T field.
(Chapter 22, Section (6) The Hall Effect - cp1e_22_#22)

ANSWER: ____________________________

9. What is the magnitude of the maximum force exerted on an aluminum rod with a 0.19-μC charge that you pass between the poles of a 0.12-T permanent magnet at a speed of $12.\,\frac{\mathrm{m}}{\mathrm{s}}$?
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_22_#7)

ANSWER: ____________________________

10. Find the current that must run through a current-carrying circular loop so that it produces a 180-μT magnetic field at its center, given that the radius of the circular loop is $17.\,\mathrm{cm}$.
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ex22_7)

ANSWER: ____________________________

11. Find the current that must run through a long, straight wire so that it produces a $11.$-μT magnetic field at a distance of $6.7\,\mathrm{cm}$ from the wire.
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ex22_6)

ANSWER: ____________________________

12. A 0.13-kg baseball, pitched at $23.\,\frac{\mathrm{m}}{\mathrm{s}}$ horizontally and perpendicularly to Earth's horizontal $52.$-μT field, has a $16.$-nC charge on it. What distance is the ball deflected from its path by the magnetic force after it travels $5.7\,\mathrm{m}$ horizontally? Do not consider the displacement due to the weight force. You may assume that the ball does not hit the ground during this motion.
(Chapter 22, Section (11) Applications of Magnetism - cp1e_22_#82)

ANSWER: ____________________________

13. Calculate the radius of curvature of the path of a proton having a velocity of $1.9 \times 10^{7}\,\mathrm{m/s}$ perpendicular to a magnetic field of strength $B = 0.21\,\mathrm{T}$.
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ex22_2)

ANSWER: ____________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question